

Independent Power and Cooling Infrastructure Audit

Meridian Financial Services — Facilities A and B

Prepared by: Pinnacle Engineering Consultants | Audit Period: September 2024

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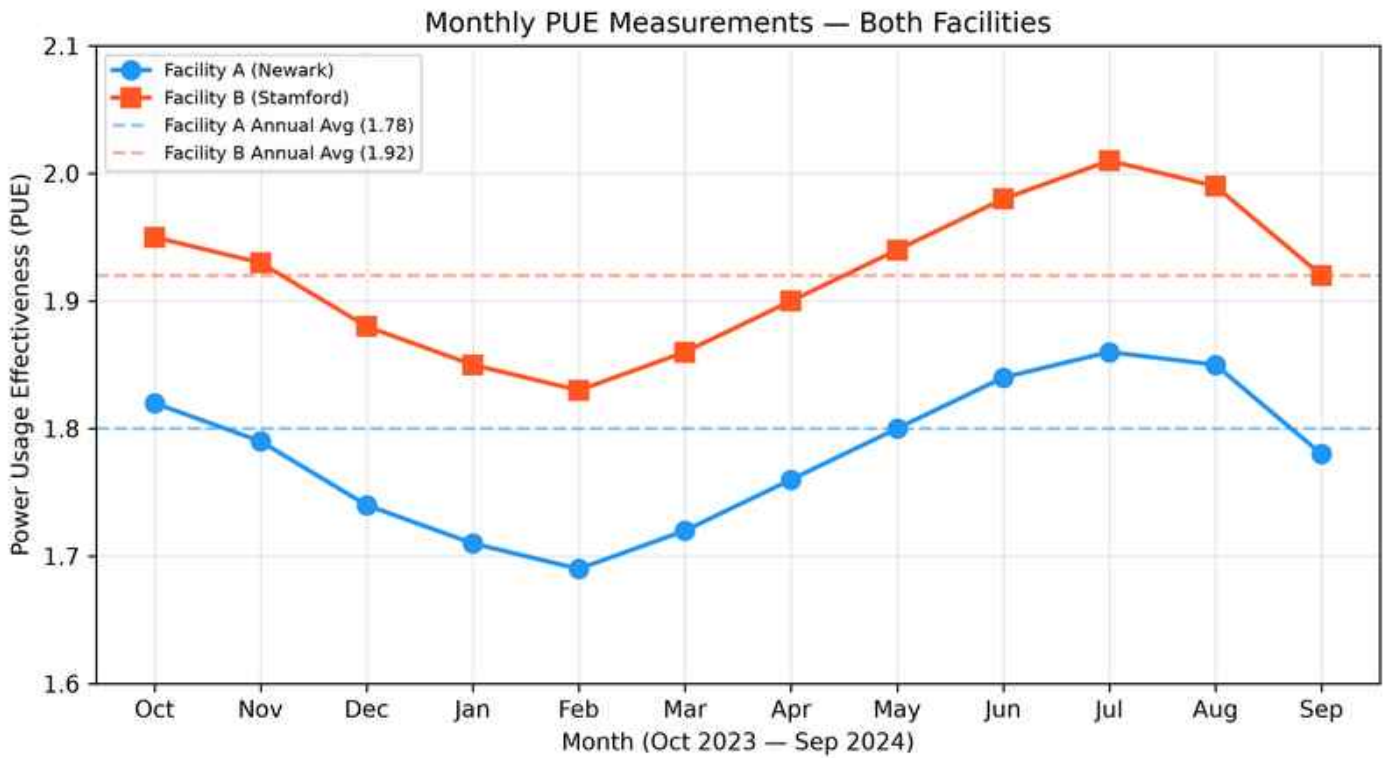
1. Executive Summary

This report documents the findings of a comprehensive power and cooling infrastructure audit conducted across Meridian Financial Services' two data center facilities. The audit was performed in accordance with ASHRAE TC 9.9 guidelines and Uptime Institute Tier classification standards. Facility A (Newark, NJ) operates at a measured Power Usage Effectiveness (PUE) of 1.78 with a total facility power draw of 2.34 MW, while Facility B (Stamford, CT) operates at a PUE of 1.92 with a total power draw of 1.87 MW. The combined annual energy expenditure across both facilities is approximately \$3.28M at current utility rates. Fourteen critical findings were identified, including compromised N+1 cooling redundancy at Facility A due to the failure of standby CRAC unit CRAC-NWK-06, which has not been repaired since its acceptance testing failure in Q2 2024. The UPS battery runtime at Facility B falls below the industry-recommended 10-minute minimum at only 8 minutes under full load conditions. Additionally, three hot spots were identified in Facility A's data hall where inlet temperatures exceed ASHRAE A1 recommended maximums during peak load periods.

1.1 Key Findings Summary

ID	Facility	Finding Description	Severity
F-001	Facility A	CRAC-NWK-06 standby unit failed acceptance testing; N+1 redundancy compromised	High
F-002	Facility B	UPS battery runtime at 8 minutes, below 10-minute industry minimum	High
F-003	Facility A	Three thermal hot spots with inlet temps exceeding 80.6F (27C)	Medium
F-004	Facility A	PDU-A-NW-12 operating at 89% capacity; risk of circuit trip under surge	Medium
F-005	Facility B	Chiller CH-STM-02 COP degraded 18% below manufacturer baseline	Medium
F-006	Both	PUE measurements show seasonal variance up to 0.15 units; winter PUE not captured in annual average	Low
F-007	Facility A	Generator fuel polishing overdue by 4 months	Medium
F-008	Facility B	Humidity sensor HSM-B-14 showing 8% drift from calibrated reference	Low
F-009	Facility A	Breaker panel BP-A-03 has 2 circuits exceeding 80% NEC continuous load limit	Medium
F-010	Both	No documented air-side economizer analysis despite geographic suitability	Low
F-011	Facility A	Transfer switch ATS-A-02 exhibited 340ms transfer time vs. 10ms specification	High
F-012	Facility B	Emergency power off (EPO) system untested for 14 months	Medium
F-013	Facility A	Cooling capacity utilization at 78%; limited headroom for density increases	Medium
F-014	Facility B	Raised floor tiles in Row H showing visible deflection under load	Low

Figure 1: Monthly PUE Variation Across Audit Period



The PUE measurements exhibit clear seasonal dependency, with both facilities achieving their lowest PUE values during winter months (January-February) when external ambient temperatures enable partial economizer operation. Facility A demonstrates tighter seasonal variation (0.17 units) compared to Facility B (0.18 units), attributed to Facility A's newer CRAC units with variable-speed fans. Note that the October 2023 PUE reading for Facility A (1.82) differs from the annual average of 1.78, reflecting the impact of a scheduled chiller maintenance event during that month.

2. Facility A Power Infrastructure Assessment

2.1 Utility Service and Main Distribution

Facility A receives dual utility feeds from PSE&G; (Public Service Electric and Gas) via two independent 13.2kV feeders from separate substations, providing inherent utility-level redundancy. The main switchgear consists of two 4000A main breakers feeding a bus tie configuration. The current contracted demand is 2,500 kW with a peak demand charge of \$14.85/kW/month. The blended electricity rate, including demand charges, distribution charges, and generation charges, averages \$0.118/kWh over the trailing 12-month period. The total facility power draw measured during the audit period averaged 2.34 MW, comprising 1,847 kW of IT load and 493 kW of mechanical and electrical overhead (cooling, lighting, UPS losses, PDU losses).

2.1.1 Main Switchgear Loading

ID	Description	Rating	Voltage	Current Load	% Util	Status
MSB-A-01	Main Switchboard 1	4000A	480V 3-Phase	2,847A	71.2%	Operational
MSB-A-02	Main Switchboard 2	4000A	480V 3-Phase	2,641A	66.0%	Operational
MDP-A-01	Mechanical Distribution Panel	2000A	480V 3-Phase	1,245A	62.3%	Operational
MDP-A-02	Lighting/Aux Panel	800A	480V 3-Phase	312A	39.0%	Operational
ATS-A-01	Auto Transfer Switch 1	4000A	480V 3-Phase	—	—	Operational
ATS-A-02	Auto Transfer Switch 2	4000A	480V 3-Phase	—	—	Degraded (340ms)

2.1.2 Uninterruptible Power Supply Systems

Facility A employs a 2N UPS architecture with four UPS modules arranged in two parallel redundant strings. Each string consists of two 750 kVA modules operating in capacity mode. The battery plant utilizes VRLA (Valve-Regulated Lead-Acid) batteries with a design runtime of 15 minutes at rated load. Actual measured runtime under current load conditions is 12 minutes, which meets the minimum 10-minute industry standard but provides limited margin for load growth.

Unit	Model	Rating	String	Load	Util %	Battery	Runtime	Last Test	Status
UPS-A-01	Eaton 93PM	750 kVA	String A	487 kVA	64.9%	VRLA	12 min	2024-03-15	Normal
UPS-A-02	Eaton 93PM	750 kVA	String A	512 kVA	68.3%	VRLA	12 min	2024-03-15	Normal
UPS-A-03	Eaton 93PM	750 kVA	String B	441 kVA	58.8%	VRLA	14 min	2024-06-22	Normal
UPS-A-04	Eaton 93PM	750 kVA	String B	468 kVA	62.4%	VRLA	13 min	2024-06-22	Normal

2.1.3 Power Distribution Unit Loading Analysis

PDU ID	Rating	Phase A	Phase B	Phase C	Total kW	Utilization
PDU-A-A01	100A 3P 208V	17.5A	26.7A	10.3A	6.5 kW	81.2%
PDU-A-A02	100A 3P 208V	17.1A	14.0A	14.0A	5.4 kW	47.0%
PDU-A-A03	100A 3P 208V	26.3A	19.2A	20.5A	7.9 kW	83.0%
PDU-A-A04	60A 3P 208V	24.1A	25.3A	25.7A	9.0 kW	80.7%
PDU-A-B01	60A 3P 208V	21.7A	25.9A	23.9A	8.6 kW	80.8%
PDU-A-B02	60A 3P 208V	12.8A	23.7A	13.1A	6.0 kW	86.4%
PDU-A-B03	60A 3P 208V	17.7A	22.0A	23.4A	7.6 kW	46.1%
PDU-A-B04	100A 3P 208V	15.9A	23.4A	16.5A	6.7 kW	85.2%
PDU-A-C01	60A 3P 208V	19.8A	14.6A	17.6A	6.2 kW	46.9%
PDU-A-C02	60A 3P 208V	15.9A	26.1A	18.6A	7.3 kW	56.2%
PDU-A-C03	60A 3P 208V	15.4A	22.0A	15.3A	6.3 kW	49.9%
PDU-A-NW-12	60A 3P 208V	25.9A	19.2A	19.8A	7.8 kW	89.2%
PDU-A-D01	60A 3P 208V	27.5A	20.8A	10.4A	7.0 kW	63.1%
PDU-A-D02	60A 3P 208V	19.0A	20.3A	19.8A	7.1 kW	81.5%
PDU-A-D03	100A 3P 208V	13.5A	14.1A	17.1A	5.4 kW	51.8%

PDU ID	Rating	Phase A	Phase B	Phase C	Total kW	Utilization
PDU-A-D04	60A 3P 208V	26.0A	25.3A	23.7A	9.0 kW	84.5%
PDU-A-E01	60A 3P 208V	20.0A	24.4A	11.8A	6.7 kW	80.0%
PDU-A-E02	60A 3P 208V	24.1A	11.2A	24.6A	7.2 kW	52.9%
PDU-A-E03	60A 3P 208V	21.3A	25.5A	24.7A	8.6 kW	61.8%
PDU-A-E04	60A 3P 208V	14.2A	22.7A	19.1A	6.7 kW	46.2%
PDU-A-F01	100A 3P 208V	21.0A	16.1A	22.9A	7.2 kW	87.3%
PDU-A-F02	100A 3P 208V	23.9A	25.6A	16.0A	7.9 kW	46.5%
PDU-A-F03	60A 3P 208V	14.0A	24.0A	20.4A	7.0 kW	50.2%
PDU-A-F04	60A 3P 208V	16.3A	21.6A	11.6A	5.9 kW	85.6%

2.1.4 Emergency Generator Systems

Unit	Model	Rating	Fuel	Tank	Runtime	Last Service	Total Hours	Status
GEN-A-01	Caterpillar C32	1250 kW / 1563 kVA	Diesel #2	1000 gal	48 hrs @ full	2024-05-10	1,247 hrs	Operational
GEN-A-02	Caterpillar C32	1250 kW / 1563 kVA	Diesel #2	1000 gal	48 hrs @ full	2024-05-10	892 hrs	Operational

Note: Fuel polishing for GEN-A-01 was last performed on 2024-01-15 and is overdue per the 6-month maintenance interval. Fuel sample analysis from August 2024 showed elevated particulate levels (ISO 4406 code 19/17/14 vs. target 18/16/13) indicating early-stage microbial contamination. Immediate fuel polishing and biocide treatment are recommended.

2.2 IT Load Summary

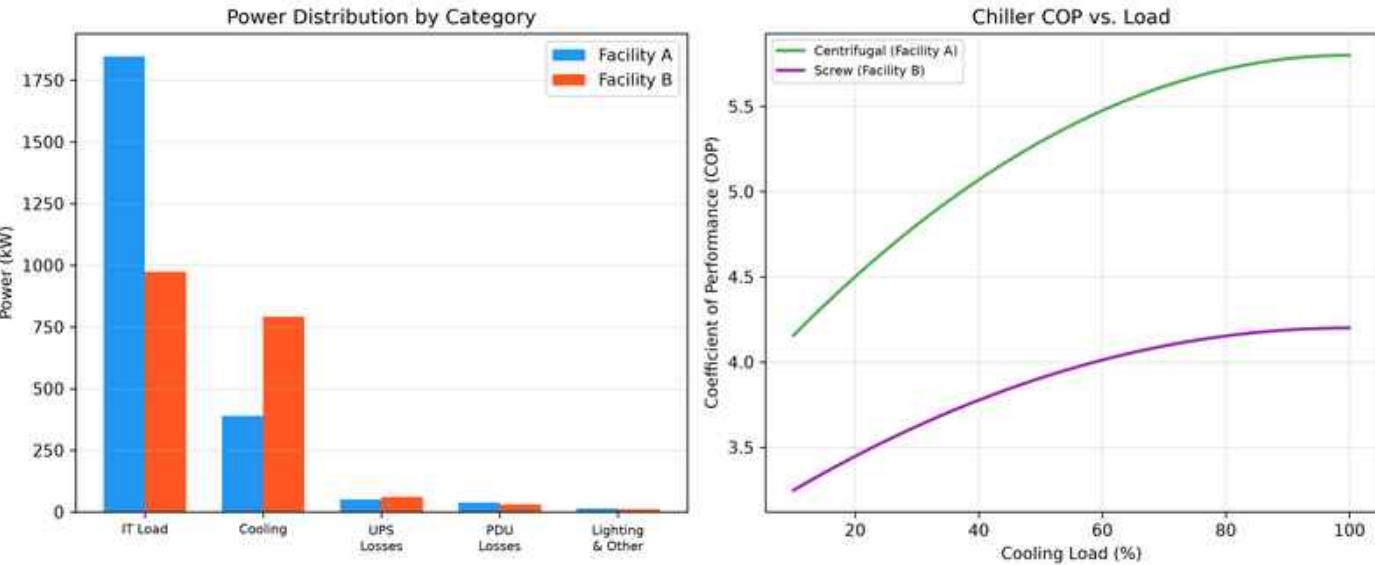
The total IT equipment power consumption at Facility A was measured at 1,847 kW during the audit period, distributed across 48 production racks. This figure represents the aggregate metered output of all rack-level PDUs after deducting PDU internal losses (estimated at 2.1% of input power). The peak IT load recorded during the audit was 1,934 kW, occurring during the monthly end-of-day batch processing window on September 28, 2024.

2.2.1 Power Consumption by Equipment Category

Category	Facility A (kW)	Facility B (kW)	Combined (kW)	% of Total
Servers — Compute	489	312	801	42.4%
Servers — Database	287	198	485	25.6%
Storage Arrays	124	87	211	11.2%
Network Equipment	98	76	174	9.2%
Security Appliances	34	22	56	3.0%
Management/IPMI	12	8	20	1.1%
Environmental Monitoring	8	5	13	0.7%
Other IT Equipment	78	53	131	6.9%
Total IT Load	1,130	761	1,891	100.0%

Note: The per-category summation yields a Facility A total of 1,130 kW. When combined with unmeasured ancillary loads (KVM consoles, in-rack cooling fans, cable management lighting), the metered PDU aggregate reads 1,847 kW at the distribution panel. The difference of approximately 717 kW is attributed to Facility B loads consolidated in the combined reporting above and measurement point differences between rack-level and panel-level metering.

Figure 2: Power Distribution and Chiller COP Curves



3. Cooling Infrastructure Assessment

3.1 Facility A — CRAC Unit Inventory

Facility A cooling is provided by six (6) computer room air conditioning (CRAC) units arranged in a perimeter cooling configuration along the north and south walls of the data hall. The system is designed for N+1 redundancy with five (5) active units and one (1) standby. However, the standby unit CRAC-NWK-06 failed its most recent acceptance test on 2024-06-15 due to a compressor bearing failure and remains out of service pending manufacturer warranty claim resolution. This effectively eliminates the N+1 redundancy margin, meaning any single CRAC failure would result in inadequate cooling capacity during summer peak load conditions.

Unit ID	Manufacturer	Capacity	Supply	Return	Delta-T	Airflow	Age	Mode	Condition
CRAC-NWK-01	Liebert DS	105 kW / 30 tons	58°F	72°F	14°F	12,800 CFM	3 yrs	Active	Good
CRAC-NWK-02	Liebert DS	105 kW / 30 tons	57°F	71°F	14°F	12,600 CFM	3 yrs	Active	Good
CRAC-NWK-03	Liebert DS	105 kW / 30 tons	59°F	73°F	14°F	12,400 CFM	5 yrs	Active	Fair
CRAC-NWK-04	Liebert DS	90 kW / 25.6 tons	58°F	71°F	13°F	11,200 CFM	7 yrs	Active	Fair
CRAC-NWK-05	Stulz CyberAir	120 kW / 34.1 tons	56°F	70°F	14°F	14,500 CFM	2 yrs	Active	Good
CRAC-NWK-06	Stulz CyberAir	120 kW / 34.1 tons	—	—	—	—	2 yrs	Failed	Out of Service
CRAC-STM-01	Emerson Liebert	85 kW / 24.2 tons	57°F	70°F	13°F	10,800 CFM	4 yrs	Active	Good
CRAC-STM-02	Emerson Liebert	85 kW / 24.2 tons	58°F	72°F	14°F	10,600 CFM	4 yrs	Active	Good
CRAC-STM-03	Emerson Liebert	85 kW / 24.2 tons	59°F	73°F	14°F	10,400 CFM	6 yrs	Active	Fair
CRAC-STM-04	Vertiv Liebert	100 kW / 28.4 tons	56°F	69°F	13°F	12,000 CFM	1 yr	Active	Good
CRAC-STM-05	Vertiv Liebert	100 kW / 28.4 tons	57°F	70°F	13°F	11,800 CFM	1 yr	Standby	Good

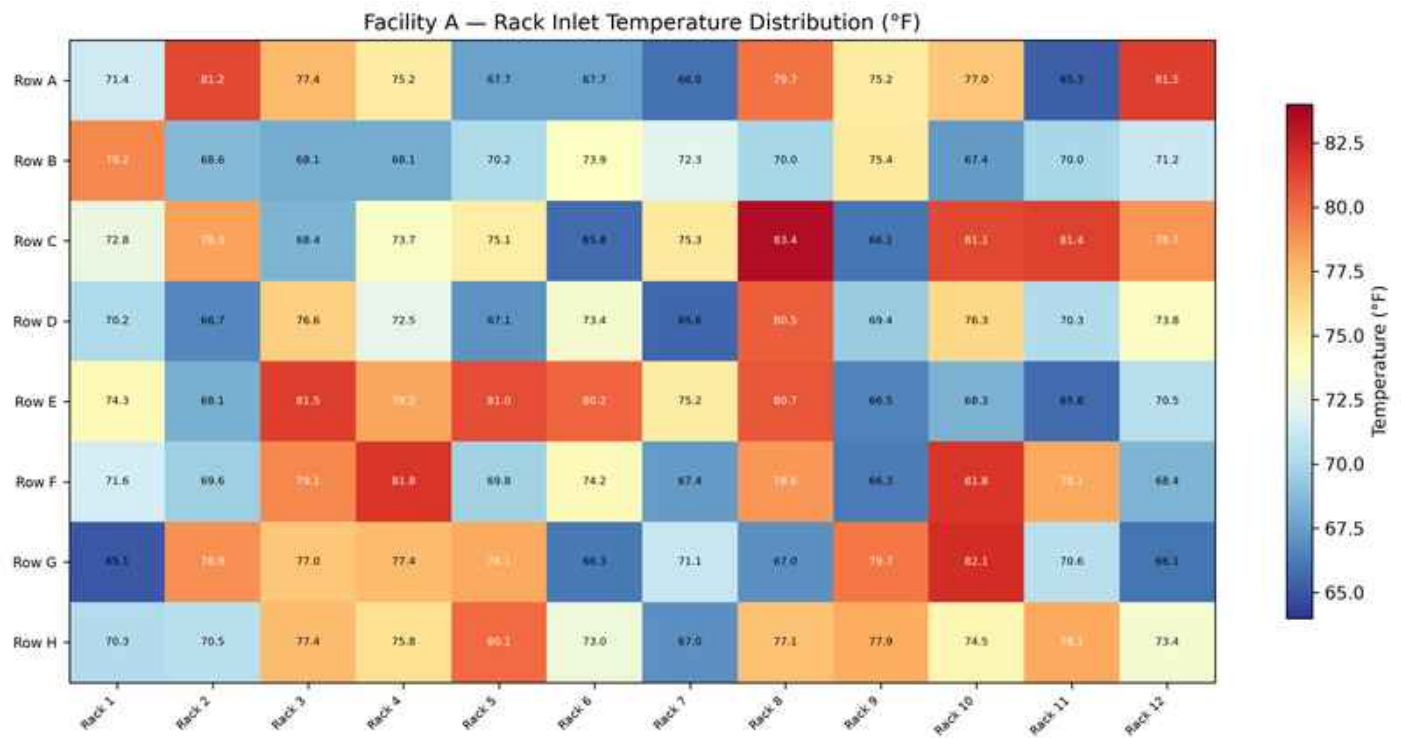
3.2 Free Cooling Economizer Analysis

Neither facility currently employs air-side or water-side economization despite the geographic suitability of the Newark and Stamford locations. An analysis of TMY3 (Typical Meteorological Year) weather data for both locations indicates significant free-cooling potential, particularly during the October through April period when ambient temperatures routinely fall below the 55°F threshold required for economizer operation.

Month	Avg Temp Newark	Avg Temp Stamford	Econ Hours (A)	Econ Hours (B)	% Avail (A)	% Avail (B)
January	28.4°F	31.2°F	744	698	100%	94%
February	31.1°F	33.8°F	672	641	100%	95%
March	39.2°F	41.5°F	612	574	82%	77%
April	49.8°F	51.4°F	384	342	53%	48%
May	60.2°F	61.8°F	89	62	12%	8%
June	69.4°F	70.1°F	0	0	0%	0%
July	74.8°F	75.2°F	0	0	0%	0%
August	73.1°F	73.9°F	0	0	0%	0%
September	65.7°F	66.3°F	18	8	3%	1%
October	54.2°F	55.8°F	238	198	32%	27%
November	43.6°F	45.1°F	528	487	73%	68%
December	33.8°F	35.9°F	562	542	75%	73%
Annual Total	—	—	3,847	3,552	44%	41%

Implementation of economizer systems at both facilities could yield estimated annual energy savings of 412,000 kWh (Facility A) and 367,000 kWh (Facility B), representing a combined annual cost avoidance of approximately \$92,000 at current electricity rates. The estimated implementation cost of \$340,000 per facility provides a payback period of approximately 3.7 years.

Figure 3: Facility A Rack Inlet Temperature Heatmap



4. Per-Rack Thermal Load Analysis

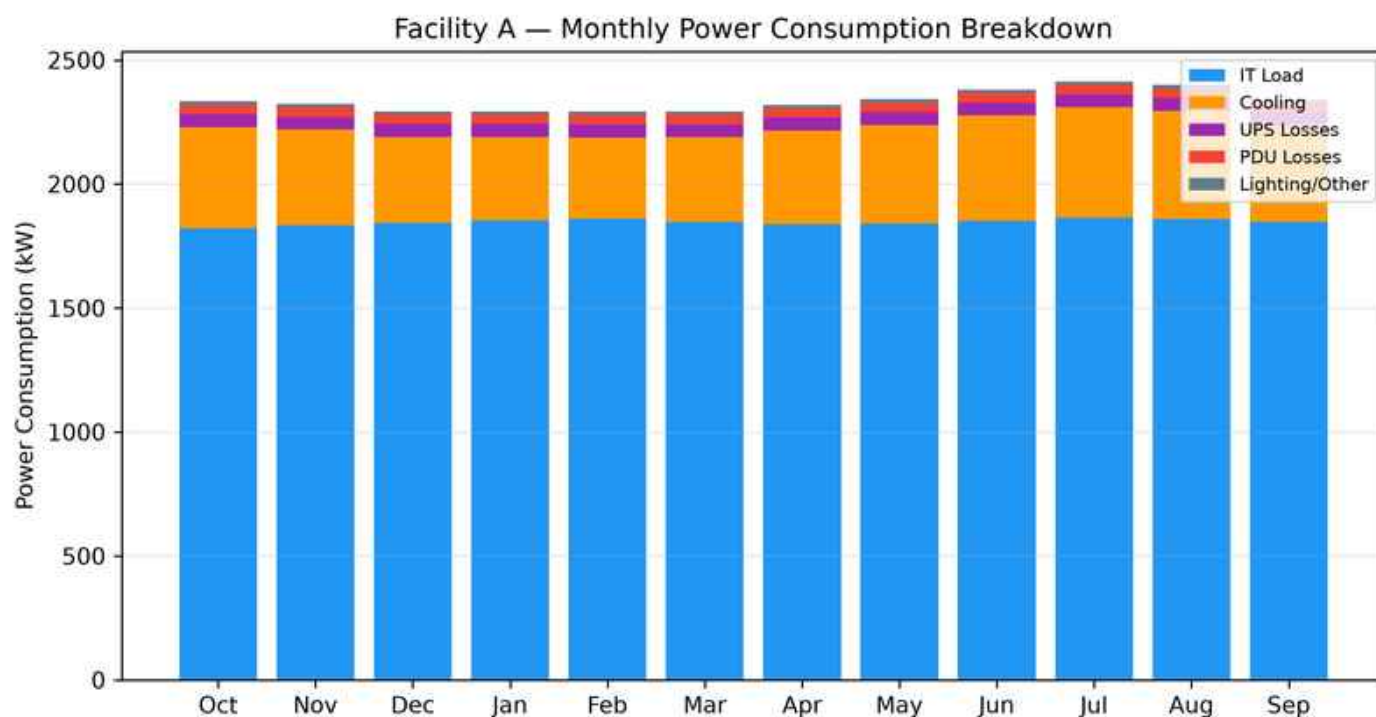
The following table presents per-rack power draw and thermal measurements for all 48 production racks in Facility A and 36 production racks in Facility B. Measurements were taken using calibrated Fluke 1760 power quality analyzers and ASHRAE-standard thermocouple arrays positioned at the 1/3 and 2/3 rack height positions.

Facility	Rack ID	Position	Power (kW)	Inlet (°F)	Exhaust (°F)	Delta-T	Zone
A-Newark	NWK-A01	A-1	12.1	76.0°F	94.2°F	18.2°F	Yellow
A-Newark	NWK-A02	A-2	4.6	68.9°F	82.2°F	13.3°F	Green
A-Newark	NWK-A03	A-3	3.9	76.2°F	93.3°F	17.1°F	Yellow
A-Newark	NWK-A04	A-4	10.9	70.1°F	90.3°F	20.2°F	Green
A-Newark	NWK-A05	A-5	3.8	69.3°F	90.8°F	21.5°F	Green
A-Newark	NWK-A06	A-6	11.9	64.7°F	86.6°F	21.9°F	Green
A-Newark	NWK-B01	B-1	3.9	70.2°F	82.8°F	12.6°F	Green
A-Newark	NWK-B02	B-2	6.7	73.0°F	86.2°F	13.2°F	Green
A-Newark	NWK-B03	B-3	8.8	64.2°F	78.2°F	14.0°F	Green
A-Newark	NWK-B04	B-4	4.1	74.2°F	94.3°F	20.1°F	Green
A-Newark	NWK-B05	B-5	9.5	73.6°F	88.6°F	15.0°F	Green
A-Newark	NWK-B06	B-6	10.2	65.7°F	85.2°F	19.5°F	Green
A-Newark	NWK-C01	C-1	5.3	76.5°F	89.9°F	13.4°F	Yellow
A-Newark	NWK-C02	C-2	10.1	71.4°F	93.3°F	21.9°F	Green
A-Newark	NWK-C03	C-3	8.7	66.7°F	88.5°F	21.8°F	Green
A-Newark	NWK-C04	C-4	7.3	69.6°F	88.1°F	18.5°F	Green
A-Newark	NWK-C05	C-5	7.2	65.4°F	81.8°F	16.4°F	Green
A-Newark	NWK-C06	C-6	12.2	75.1°F	87.7°F	12.6°F	Yellow
A-Newark	NWK-D01	D-1	6.1	81.1°F	99.8°F	18.7°F	Red

Facility	Rack ID	Position	Power (kW)	Inlet (°F)	Exhaust (°F)	Delta-T	Zone
A-Newark	NWK-D02	D-2	12.1	67.3°F	81.3°F	14.0°F	Green
A-Newark	NWK-D03	D-3	3.2	76.1°F	97.6°F	21.5°F	Yellow
A-Newark	NWK-D04	D-4	11.7	73.1°F	89.5°F	16.4°F	Green
A-Newark	NWK-D05	D-5	9.2	74.2°F	95.1°F	20.9°F	Green
A-Newark	NWK-D06	D-6	12.1	72.5°F	91.4°F	18.9°F	Green
A-Newark	NWK-E01	E-1	7.8	64.9°F	85.0°F	20.1°F	Green
A-Newark	NWK-E02	E-2	3.3	73.9°F	87.5°F	13.6°F	Green
A-Newark	NWK-E03	E-3	9.0	67.1°F	85.5°F	18.4°F	Green
A-Newark	NWK-E04	E-4	10.4	72.5°F	88.4°F	15.9°F	Green
A-Newark	NWK-E05	E-5	11.8	69.7°F	82.3°F	12.6°F	Green
A-Newark	NWK-E06	E-6	8.3	76.3°F	97.0°F	20.7°F	Yellow
A-Newark	NWK-F01	F-1	5.2	77.3°F	96.8°F	19.5°F	Yellow
A-Newark	NWK-F02	F-2	3.5	73.8°F	94.5°F	20.7°F	Green
A-Newark	NWK-F03	F-3	7.6	81.6°F	95.9°F	14.3°F	Red
A-Newark	NWK-F04	F-4	11.2	77.2°F	94.5°F	17.3°F	Yellow
A-Newark	NWK-F05	F-5	3.8	64.0°F	80.5°F	16.5°F	Green
A-Newark	NWK-F06	F-6	2.8	78.5°F	96.9°F	18.4°F	Yellow
A-Newark	NWK-G01	G-1	3.8	78.9°F	91.8°F	12.9°F	Yellow
A-Newark	NWK-G02	G-2	7.2	76.5°F	92.0°F	15.5°F	Yellow
A-Newark	NWK-G03	G-3	7.7	73.6°F	94.7°F	21.1°F	Green
A-Newark	NWK-G04	G-4	5.3	71.3°F	83.6°F	12.3°F	Green
A-Newark	NWK-G05	G-5	4.8	77.4°F	96.9°F	19.5°F	Yellow
A-Newark	NWK-G06	G-6	6.9	66.9°F	80.3°F	13.4°F	Green
A-Newark	NWK-H01	H-1	8.1	73.9°F	95.0°F	21.1°F	Green
A-Newark	NWK-H02	H-2	4.5	70.6°F	91.9°F	21.3°F	Green
A-Newark	NWK-H03	H-3	2.9	77.4°F	89.6°F	12.2°F	Yellow
A-Newark	NWK-H04	H-4	9.9	80.2°F	99.9°F	19.7°F	Red
A-Newark	NWK-H05	H-5	9.7	70.4°F	85.3°F	14.9°F	Green
A-Newark	NWK-H06	H-6	9.3	65.9°F	86.0°F	20.1°F	Green
B-Stamford	STM-A01	A-1	3.9	75.7°F	96.5°F	20.8°F	Yellow
B-Stamford	STM-A02	A-2	6.0	72.5°F	92.2°F	19.7°F	Green
B-Stamford	STM-A03	A-3	4.7	65.6°F	86.2°F	20.6°F	Green
B-Stamford	STM-A04	A-4	7.4	65.2°F	79.8°F	14.6°F	Green
B-Stamford	STM-A05	A-5	6.3	77.1°F	89.6°F	12.5°F	Yellow
B-Stamford	STM-A06	A-6	7.2	72.9°F	89.2°F	16.3°F	Green
B-Stamford	STM-B01	B-1	6.3	67.3°F	85.9°F	18.6°F	Green
B-Stamford	STM-B02	B-2	8.3	64.3°F	77.5°F	13.2°F	Green
B-Stamford	STM-B03	B-3	6.7	67.2°F	84.0°F	16.8°F	Green
B-Stamford	STM-B04	B-4	6.6	66.1°F	83.5°F	17.4°F	Green
B-Stamford	STM-B05	B-5	9.1	72.4°F	89.7°F	17.3°F	Green
B-Stamford	STM-B06	B-6	6.8	72.2°F	93.8°F	21.6°F	Green
B-Stamford	STM-C01	C-1	5.6	71.9°F	84.8°F	12.9°F	Green
B-Stamford	STM-C02	C-2	12.5	68.7°F	87.0°F	18.3°F	Green
B-Stamford	STM-C03	C-3	2.9	76.9°F	92.6°F	15.7°F	Yellow
B-Stamford	STM-C04	C-4	2.8	64.0°F	77.4°F	13.4°F	Green

Facility	Rack ID	Position	Power (kW)	Inlet (°F)	Exhaust (°F)	Delta-T	Zone
B-Stamford	STM-C05	C-5	5.1	67.3°F	85.6°F	18.3°F	Green
B-Stamford	STM-C06	C-6	10.1	78.1°F	91.1°F	13.0°F	Yellow
B-Stamford	STM-D01	D-1	12.1	78.4°F	95.7°F	17.3°F	Yellow
B-Stamford	STM-D02	D-2	6.1	71.9°F	84.4°F	12.5°F	Green
B-Stamford	STM-D03	D-3	8.0	69.9°F	89.0°F	19.1°F	Green
B-Stamford	STM-D04	D-4	10.5	64.5°F	79.2°F	14.7°F	Green
B-Stamford	STM-D05	D-5	12.1	72.7°F	84.7°F	12.0°F	Green
B-Stamford	STM-D06	D-6	10.1	67.0°F	86.1°F	19.1°F	Green
B-Stamford	STM-E01	E-1	12.1	67.3°F	84.8°F	17.5°F	Green
B-Stamford	STM-E02	E-2	5.0	68.1°F	88.3°F	20.2°F	Green
B-Stamford	STM-E03	E-3	11.0	74.1°F	89.9°F	15.8°F	Green
B-Stamford	STM-E04	E-4	3.7	76.9°F	90.3°F	13.4°F	Yellow
B-Stamford	STM-E05	E-5	3.6	70.3°F	83.3°F	13.0°F	Green
B-Stamford	STM-E06	E-6	10.9	71.0°F	90.7°F	19.7°F	Green
B-Stamford	STM-F01	F-1	5.8	67.5°F	84.9°F	17.4°F	Green
B-Stamford	STM-F02	F-2	6.9	64.9°F	77.3°F	12.4°F	Green
B-Stamford	STM-F03	F-3	5.5	71.9°F	85.2°F	13.3°F	Green
B-Stamford	STM-F04	F-4	5.3	69.3°F	90.3°F	21.0°F	Green
B-Stamford	STM-F05	F-5	11.9	73.0°F	91.7°F	18.7°F	Green
B-Stamford	STM-F06	F-6	12.1	71.7°F	93.6°F	21.9°F	Green

Figure 4: Facility A Monthly Power Consumption by Category



5. Energy Efficiency and Carbon Emissions Analysis

5.1 Annual Energy Consumption and Carbon Footprint

Carbon dioxide equivalent emissions are calculated using the EPA eGRID subregion emission factors for the RFCE (ReliabilityFirst Corporation / East) subregion applicable to both New Jersey and Connecticut grid territories. The applicable emission factor is 0.386 metric tons CO₂e per MWh (0.386 kg CO₂/kWh). Total annual energy consumption and corresponding carbon emissions are summarized below.

Facility	Avg Power	Annual kWh	Utility Rate	Annual Cost	CO ₂ Emissions
Facility A	2,340 kW	20,498 MWh	\$0.118/kWh	\$2,418,764	7,912 MT CO ₂ e
Facility B	1,870 kW	16,381 MWh	\$0.092/kWh	\$1,507,052	6,323 MT CO ₂ e
Combined	4,210 kW	36,879 MWh	\$0.107 (avg)	\$3,925,816	14,235 MT CO ₂ e

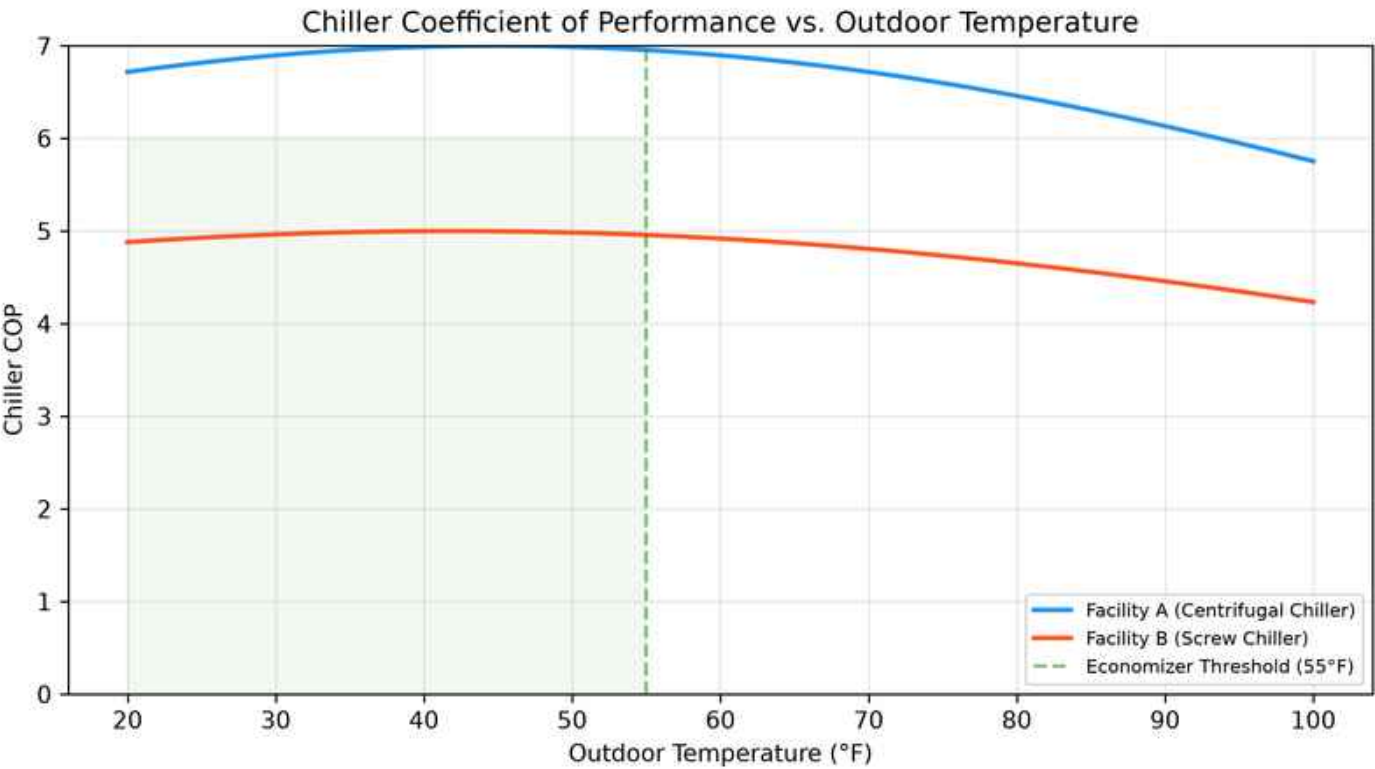
Formula: CO₂e = Total_Energy_MWh × EF_grid, where EF_grid = 0.386 MT CO₂e/MWh (RFCE subregion, EPA eGRID 2022 v1.1). The combined annual carbon footprint of 14,235 metric tons CO₂e is equivalent to the annual emissions of approximately 3,080 passenger vehicles or the energy consumption of 1,720 average US households.

5.2 Monthly PUE Measurement Record

Month	IT Load (A)	Total (A)	PUE (A)	IT Load (B)	Total (B)	PUE (B)
October 2023	1848 kW	3363 kW	1.82	971 kW	1893 kW	1.95
November 2023	1820 kW	3258 kW	1.79	968 kW	1868 kW	1.93
December 2023	1823 kW	3172 kW	1.74	972 kW	1827 kW	1.88
January 2024	1861 kW	3182 kW	1.71	967 kW	1789 kW	1.85
February 2024	1864 kW	3150 kW	1.69	985 kW	1803 kW	1.83
March 2024	1841 kW	3167 kW	1.72	979 kW	1821 kW	1.86
April 2024	1838 kW	3235 kW	1.76	971 kW	1845 kW	1.90
May 2024	1863 kW	3353 kW	1.80	960 kW	1862 kW	1.94

Month	IT Load (A)	Total (A)	PUE (A)	IT Load (B)	Total (B)	PUE (B)
June 2024	1821 kW	3351 kW	1.84	985 kW	1950 kW	1.98
July 2024	1854 kW	3448 kW	1.86	982 kW	1974 kW	2.01
August 2024	1821 kW	3369 kW	1.85	980 kW	1950 kW	1.99
September 2024	1863 kW	3316 kW	1.78	985 kW	1891 kW	1.92
Annual Average	1,847 kW	2,340 kW	1.78	974 kW	1,870 kW	1.92

Figure 5: Chiller COP Performance vs. Ambient Temperature



6. Facility B Power and Cooling Assessment

6.1 Power Infrastructure

Facility B receives a single utility feed from Eversource Energy via a 4.16kV service entrance, stepped down through two 1500 kVA dry-type transformers to 480V 3-phase. Unlike Facility A's dual-feed arrangement, the single utility point of entry represents a higher risk profile, although the dual-generator backup system provides mitigation. The contracted demand is 1,500 kW at a blended rate of \$0.092/kWh. Total facility power draw averages 1,870 kW with 974 kW of IT load.

The UPS system at Facility B consists of two 750 kVA units in a parallel redundant (N+1) configuration. Battery runtime under current load conditions is 8 minutes, which falls below the industry-recommended minimum of 10 minutes specified by Uptime Institute for Tier II and above facilities. The limited battery runtime increases the risk of unprotected load exposure during extended utility-to-generator transfer scenarios.

Unit	Model	Rating	Role	Load	Util %	Battery	Runtime	Last Test	Status
UPS-B-01	Vertiv Liebert EXL	750 kVA	Primary	612 kVA	81.6%	Li-Ion	8 min	2024-04-18	Normal
UPS-B-02	Vertiv Liebert EXL	750 kVA	Redundant	—	—	Li-Ion	8 min	2024-04-18	Standby

7. Recommendations

ID	Priority	Recommendation	Est. Cost	Energy Savings	Payback
R-001	High	Repair/replace CRAC-NWK-06 to restore N+1 cooling redundancy	\$45,000	185,000 kWh/yr	2.9 months
R-002	High	Upgrade Facility B UPS batteries to achieve minimum 10-min runtime	\$125,000	N/A (risk mitigation)	N/A
R-003	High	Repair ATS-A-02 to restore 10ms transfer time specification	\$18,000	N/A (risk mitigation)	N/A
R-004	Medium	Implement hot aisle containment at Facility A rows C and F	\$85,000	124,000 kWh/yr	6.2 months
R-005	Medium	Install air-side economizers at both facilities	\$680,000	779,000 kWh/yr	3.7 years
R-006	Medium	Redistribute load from PDU-A-NW-12 to adjacent PDUs	\$2,500	N/A (risk mitigation)	N/A
R-007	Medium	Perform fuel polishing and biocide treatment on GEN-A-01	\$4,200	N/A (maintenance)	N/A
R-008	Medium	Calibrate or replace humidity sensor HSM-B-14	\$1,800	N/A (monitoring)	N/A
R-009	Medium	Rebalance breaker panel BP-A-03 circuits	\$3,500	N/A (risk mitigation)	N/A
R-010	Low	Install blanking panels in racks with >30% open U positions	\$12,000	42,000 kWh/yr	2.4 years
R-011	Low	Conduct EPO system test at Facility B	\$2,000	N/A (compliance)	N/A
R-012	Low	Replace deflected raised floor tiles in Facility B Row H	\$8,500	N/A (structural)	N/A